EE1003
Introduction to Computer I

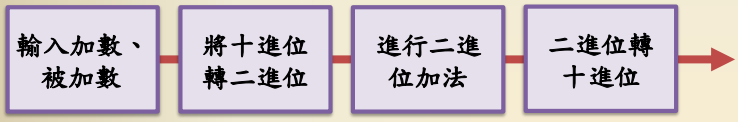
PA2
Solution

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2021.11.25

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PA2

◆使用2進位加法器的方式進行運算



PA2

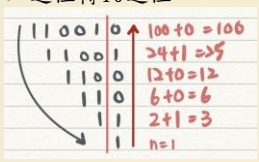
```
long long int tentotwo(int n){
    if(n<2){
        return n;
    }
    else{
        return tentotwo(n/2)*10+n%2;
    }
}
```

```
long long int twototen(long long int n){
    if(n<2){
        return n;
    }
    else{
        return twototen(n/10)*2+n%2;
    }
}
```

- 十進位轉二進位



- 二進位轉10進位



PA2

```
long long int add(long long int n1,long long int n2,long long int c){
    if(n1!=0 || n2!=0){
        return add(n1/10,n2/10,(n1%2+n2%2+c)/2)*10+(n1%2+n2%2+c)%2;
    }
    else{
        return c;
    }
}
```

- n1為被加數、n2為加數、c為進位值
- 從個位數開始加起

